Synthetic Semantic Dynamics Under Brand Coherence Field Theory: A Simulation-Based Technical Report

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Abstract

This technical report provides the computational companion to *Brand Coherence Field Theory* (BCFT), a framework that models brand meaning as a family of probability distributions over a semantic space and defines coherence as divergence from an ideal brand distribution (Abreu, 2026). Whereas the conceptual paper develops the theory and a simulation *design*, this report supplies the formal apparatus and the *executed* experiments: explicit divergences, Fisher–Rao geodesics, and natural-gradient coherence flows on the probability simplex, together with a fully reproducible Python pipeline and a synthetic dataset. We report three results. First, coherence declines monotonically with controlled distortion, and distortion of *core* semantic dimensions is roughly 35× more costly than distortion of peripheral dimensions. Second, Fisher–Rao geodesics recover the intrinsic shortest repositioning path exactly, while Euclidean interpolation is longer and strays measurably from it. Third, the natural-gradient coherence flow converges geometrically to the ideal distribution from all tested initial states. The report fixes the canonical definition of BCFT verbatim so that the conceptual and computational strands remain locked together. All code, data, and figures are released for replication.

Keywords: brand coherence; information geometry; Fisher–Rao metric; Kullback–Leibler divergence; natural gradient; statistical manifold; reproducible simulation.

1 Introduction and scope

Brand Coherence Field Theory (BCFT) was introduced as a conceptual framework in the companion working paper (Abreu, 2026), where the brand is reconceived not as a fixed bundle of attributes but as a structured field of semantic probability distributions, and coherence is defined as (negative) divergence from an ideal brand distribution. That paper deliberately presents the simulation at the level of *design* and *expected behaviour*, in keeping with its conceptual contribution and its intended marketing-science audience.

This report is the technical counterpart. Its purpose is narrow and explicit: to *execute* the BCFT simulation programme with real computation, so that the formal claims of the theory can be inspected, falsified, and reused. We make no new empirical claims about real brands here; the data are synthetic by construction, which is precisely what makes the behaviour of the formalism transparent. The report contributes (i) a self-contained mathematical formalization of BCFT on the probability simplex, (ii) reference implementations of the divergence, geodesic, and natural-gradient

machinery, (iii) three reproducible experiments with real figures and numbers, and (iv) a synthetic dataset and complete pipeline released for replication.

To keep the conceptual and computational strands of BCFT from drifting apart, the next section reproduces the theory's canonical definition verbatim before we formalize it.

2 The canonical definition of BCFT

The following statement is reproduced *verbatim* from the companion working paper (Abreu, 2026, §3); the present report adopts it without modification.

Canonical definition. Brand Coherence Field Theory (BCFT) rests on four primitives, stated here once and used without modification throughout this paper. Companion works may develop other formulations of BCFT — for example a topological–dynamical formulation over a carrier space — which share this definition as their information-geometric special case. (i) *Semantic space*: a brand's possible meanings are represented in a semantic space, instantiated empirically as expert-coded categories, topic proportions, or embeddings projected onto the probability simplex. (ii) *Brand state*: at each observation point o — a channel, campaign, stakeholder group, or moment — the realised meaning of the brand is a probability distribution P_o over the semantic space. (iii) *Ideal distribution*: the brand's intended identity is a reference distribution P^* over the same space, estimated from core strategic materials. (iv) *Coherence*: the coherence of an observed state is $C(o) = -D(P_o \,\|\, P^*)$, the negative divergence of the observed state from the ideal, where D is a divergence such as the Kullback–Leibler or Jensen–Shannon divergence. The brand is then a field $\mathcal{F}_B = \{P_o : o \in \mathcal{O}\}$ of such states over the set $\mathcal{O}$ of relevant observation points, and coherence is a graded, relational, and measurable property of that field rather than a binary attribute of the brand as a whole. In the general three-representation formulation of BCFT, this reference is the projected identity, and $C(o)$ is the single-reference reduction of a pairwise-alignment field among projected, perceived, and experienced identities.

3 Mathematical formalization

3.1 Semantic space and brand states

Fix a finite semantic space with K dimensions. A brand state is a point of the $(K-1)$-dimensional probability simplex

$$\Delta^{K-1} = \Big\{ p \in \mathbb{R}^K : p_i \geq 0,\ \textstyle\sum_{i=1}^K p_i = 1 \Big\}. \tag{1}$$

An observation point $o \in \mathcal{O}$ (channel, campaign, stakeholder group, moment) yields a brand state $P_o \in \Delta^{K-1}$; the ideal distribution $P^* \in \Delta^{K-1}$ encodes the intended identity.

3.2 Coherence as divergence

For a divergence $D : \Delta^{K-1} \times \Delta^{K-1} \to [0, \infty)$, BCFT coherence is

$$C(o) = -\,D(P_o \,\|\, P^*)\,. \tag{2}$$

We use two divergences. The Kullback–Leibler divergence,

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_{i=1}^{K} p_i \log \frac{p_i}{q_i}, \tag{3}$$

and the Jensen–Shannon divergence, $\mathrm{JS}(p,q) = \frac{1}{2}D_{\mathrm{KL}}(p \,\|\, m) + \frac{1}{2}D_{\mathrm{KL}}(q \,\|\, m)$ with $m = \frac{1}{2}(p+q)$, which is symmetric and bounded by $\log 2$. Both belong to the family of f-divergences (Csiszár, 1967), and their local second-order structure is the Fisher information metric below (Amari & Nagaoka, 2000).

3.3 The brand field

The brand field is the indexed collection

$$\mathcal{F}_B = \{P_o : o \in \mathcal{O}\}, \tag{4}$$

so that incoherence can be localized (one channel), clustered (one stakeholder segment), or diffuse across $\mathcal{O}$.

3.4 Fisher–Rao geometry

The simplex carries the Fisher–Rao metric, the unique Riemannian metric (up to scale) invariant under sufficient statistics (Rao, 1945; Chentsov, 1982). Under the diffeomorphism $s = \sqrt{p}$ (coordinatewise), Δ^{K-1} maps onto the positive orthant of the unit sphere S^{K-1}, where geodesics are great-circle arcs. The induced geodesic distance is

$$d_{\mathrm{FR}}(p,q) = 2\arccos\Big(\textstyle\sum_{i=1}^{K}\sqrt{p_i q_i}\Big), \tag{5}$$

the argument being the Bhattacharyya coefficient. The Fisher–Rao geodesic from p to q is obtained by spherical interpolation of $s = \sqrt{p}$,

$$\gamma(t) = \left(\frac{\sin((1-t)\omega)\,\sqrt{p} + \sin(t\omega)\,\sqrt{q}}{\sin\omega}\right)^2, \quad \omega = \arccos\big(\textstyle\sum_i \sqrt{p_i q_i}\big), \tag{6}$$

for $t \in [0,1]$, against which we compare the naive Euclidean interpolation $p^{\mathrm{euc}}(t) = (1-t)p + tq$.

3.5 Natural-gradient coherence flow

Coherence-seeking adaptation follows the natural gradient of the divergence with respect to the Fisher metric (Amari, 2016). On the simplex this yields the replicator / mirror-descent flow

$$\frac{dp_i}{dt} = -\eta\, p_i(g_i - \langle p, g\rangle), \qquad g = \nabla_p D(p \,\|\, P^*)\,, \tag{7}$$

with step size $\eta > 0$. For $D = D_{\mathrm{KL}}$ one has $g_i = \log(p_i/P_i^*) + 1$, and the flow keeps p on the simplex and decreases $D_{\mathrm{KL}}(p \,\|\, P^*)$ monotonically toward its minimum at $p = P^*$. The discrete update used in the experiments is $p_i \leftarrow p_i \exp(-\eta(g_i - \langle p, g\rangle))$, renormalized.

Proposition 1 (Fixed point)**.** *P^* is the unique fixed point of (7) in the interior of Δ^{K-1} for $D = D_{\mathrm{KL}}$, and $D_{\mathrm{KL}}(p(t) \,\|\, P^*)$ is non-increasing along the flow.*

The experiments in §5.3 confirm this behaviour numerically (geometric convergence to machine precision).

4 Synthetic experimental setup

We instantiate the running example of the companion paper: a mission-led outdoor brand with $K = 6$ semantic dimensions — *durability*, *environmental stewardship*, *repairability*, *functional competence*, *style/aesthetics*, and *status/luxury*. The first four are *core* (mission/technical); the last two are *peripheral*. The ideal distribution concentrates mass on the core:

$$P^* = (0.24,\ 0.30,\ 0.20,\ 0.18,\ 0.05,\ 0.03).$$

Observed states are generated by a controlled perturbation operator that shifts a fraction (the *perturbation magnitude*) of mass off a chosen set of dimensions and reallocates it over the complement using a Dirichlet weighting. All randomness is seeded (`SEED = 20260628`); the pipeline reproduces every number and figure below exactly.

5 Results

5.1 Experiment 1 — divergence and incoherence

We vary the perturbation magnitude from 0 to 0.95 and measure divergence from P^*, averaging over 200 replications per level, separately for core and peripheral perturbations. Divergence rises monotonically with distortion, and coherence (2) declines correspondingly (Figure 1). Crucially, the cost of distortion is structurally weighted: at maximum magnitude, core distortion produces $D_{\mathrm{KL}}(P_o \,\|\, P^*) \approx 2.40$ against ≈ 0.069 for peripheral distortion — a ratio of about $\mathbf{34.8}\times$. BCFT therefore captures that not all deviations are equally damaging; movement away from core meanings is far more incoherent than equivalent movement in peripheral style cues.

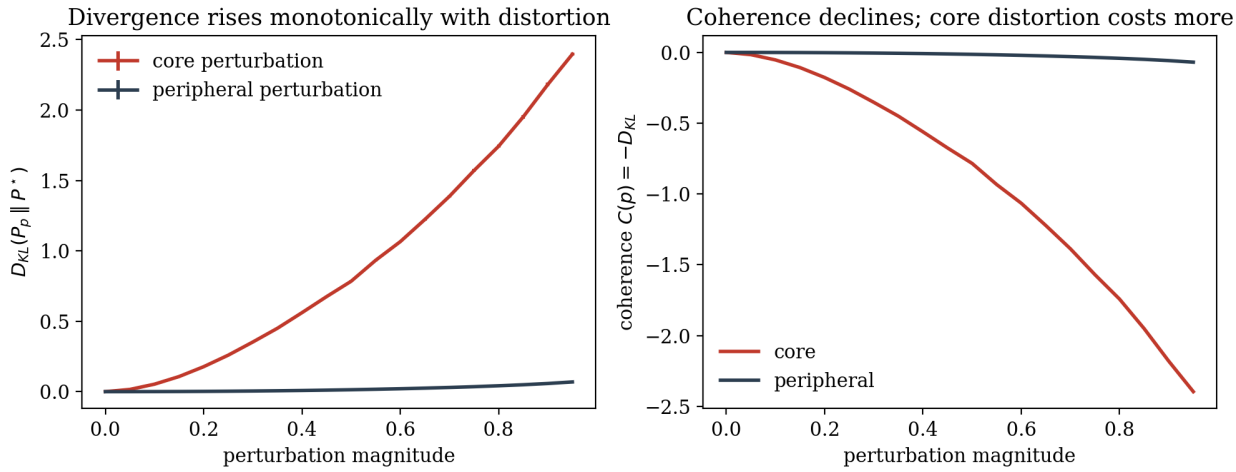


Figure 1: Experiment 1. Left: Kullback–Leibler divergence from the ideal rises monotonically with perturbation magnitude, steeply for core distortion and weakly for peripheral distortion. Right: the corresponding coherence $C(o) = -D_{\mathrm{KL}}$. Error bars are standard errors over 200 replications.

5.2 Experiment 2 — geodesic versus Euclidean movement

We compare repositioning from a core-heavy state $(0.80, 0.15, 0.05)$ to a status-heavy state $(0.05, 0.15, 0.80)$ on the $K = 3$ simplex, under the Fisher–Rao geodesic (6) and under Euclidean interpolation (Figure 2). The geodesic length equals the Fisher–Rao distance (5) to numerical precision (1.9769 vs.

1.9769), validating the implementation. The Euclidean path is longer (1.9930, an excess of **0.81**%) and, more importantly, traverses *different* intermediate brand states: the maximum Fisher–Rao deviation between the two paths at equal interpolation parameter is **0.120**. The shortest path in a flat projection is thus not the intrinsically shortest — nor the same — trajectory of semantic change, which is the strategic intuition BCFT formalizes.

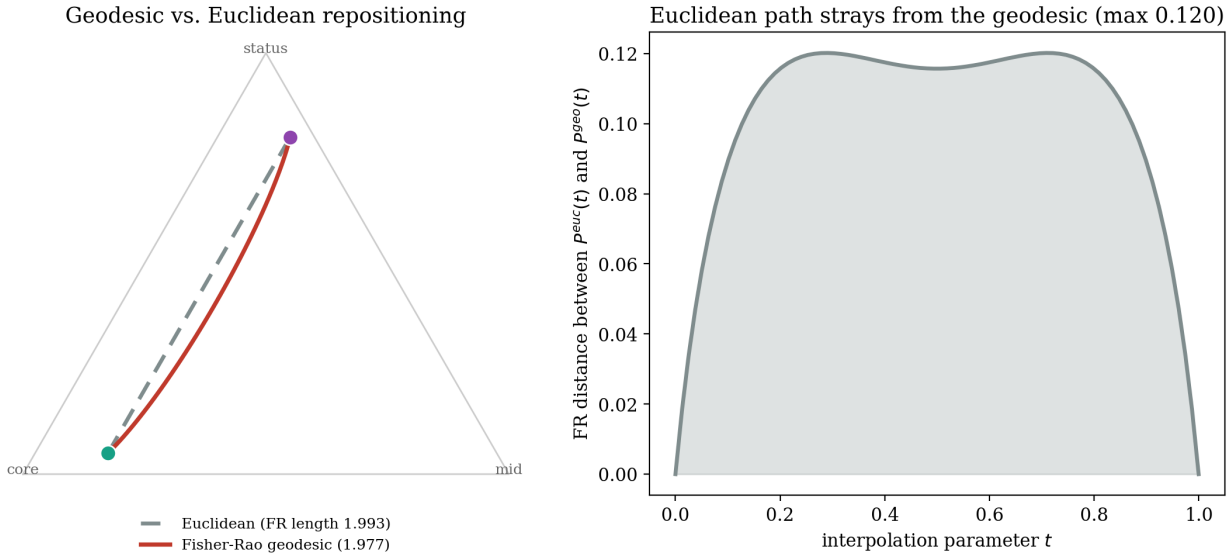


Figure 2: Experiment 2. Left: on the 3-dimensional simplex, the Fisher–Rao geodesic (solid) bows away from the Euclidean chord (dashed). Right: the Fisher–Rao distance between the Euclidean and geodesic states at the same interpolation parameter t, peaking at 0.120.

5.3 Experiment 3 — convergent coherence flows

We initialise the natural-gradient flow (7) from six random states and run it toward P^* with $\eta = 0.25$ for 150 iterations (Figure 3). All trajectories converge geometrically; the largest final divergence across runs is $\approx 1.4 \times 10^{-16}$ (machine precision), confirming the fixed-point proposition. A $K = 3$ projection shows the trajectories descending onto the ideal. This operationalizes BCFT's notion of *semantic repair*: corrective communication that follows the geometry of the semantic space restores coherence efficiently.

5.4 Synthetic brand-field dataset

For downstream replication we release a synthetic brand field of 100 observations across four source types (institutional, campaign, reviews, lifestyle), each generated by perturbing P^* with source-specific magnitude and recording the full distribution together with its KL, JS, and Fisher–Rao distances to the ideal and its coherence. Table 1 collects the headline quantities of all three experiments.

6 Discussion and limitations

The experiments demonstrate the internal coherence and computational tractability of BCFT, not its empirical validity. Three limitations follow directly from the synthetic design. First, the

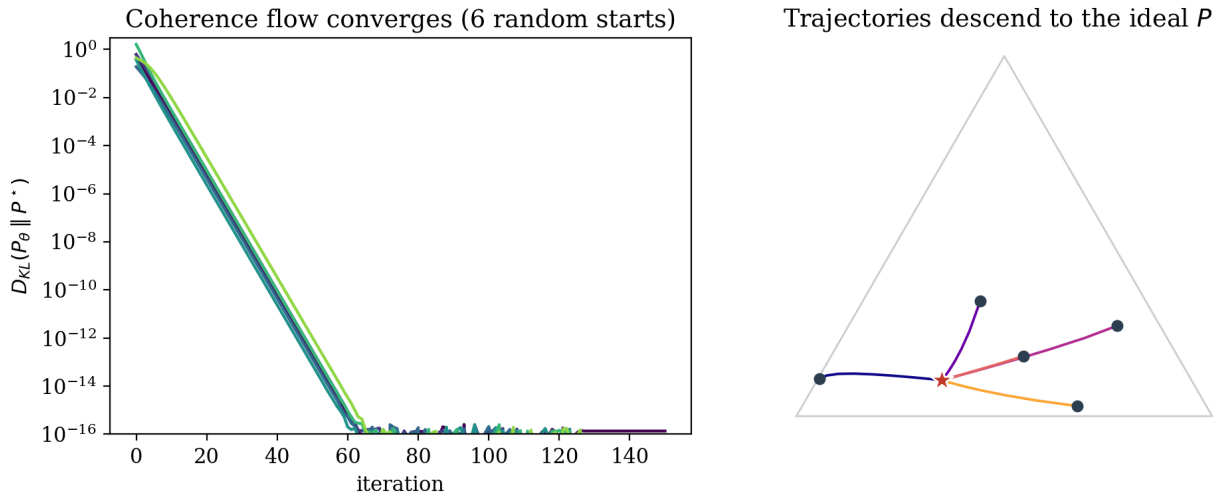


Figure 3: Experiment 3. Left: $D_{\mathrm{KL}}(P_\theta \| P^*)$ on a log scale for six random initializations; all converge to machine precision. Right: a ternary projection of trajectories descending to the ideal P^* (star).

Table 1: Headline results (seed 20260628).

Experiment	Quantity	Value
1	KL monotonic in distortion (core)	yes
1	KL at max distortion, core	2.396
1	KL at max distortion, peripheral	0.069
1	core/peripheral cost ratio	34.8×
2	Fisher–Rao distance (endpoints)	1.9769
2	geodesic length = FR distance	yes (1.9769)
2	Euclidean path length	1.9930 (+0.81%)
2	max FR deviation (Euclidean vs geodesic)	0.120
3	natural-gradient flow converges (6 starts)	yes
3	max final D_{KL}	1.4×10^{-16}

semantic space here is stipulated; in application the choice of dimensions and the estimation of P^* are substantive modelling decisions that will shape coherence profiles. Second, divergence choice matters: KL and JS encode different sensitivities, and results may shift under other f-divergences or under the Wasserstein geometry, which is non-invariant but metric on the sample space (Amari, 2021). Third, the natural-gradient flow is shown to converge for the convex KL objective; richer objectives — for example multi-ideal or mixture coherence — may exhibit local traps, which we leave to future work. These constructions are now computationally accessible through general-purpose information-geometry tooling (Le Brigant et al., 2023), and the present pipeline is designed to extend to them. The empirical programme — estimating P_o from real brand text along the lines of computational brand-image work (Mitra & Jenamani, 2020; Ranfagni et al., 2021) — is the subject of the separate empirical paper.

Reproducibility statement

All results are produced by the released pipeline. The core library (`src/bcft.py`) implements the divergences, Fisher–Rao distance, geodesic, and natural-gradient flow; `src/run_experiments.py`

regenerates every figure, the synthetic dataset, and the results summary from a single fixed seed. See `README.md` for environment and execution instructions; `requirements.txt` pins dependencies. The repository is intended for archival on Zenodo with a citable DOI.

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